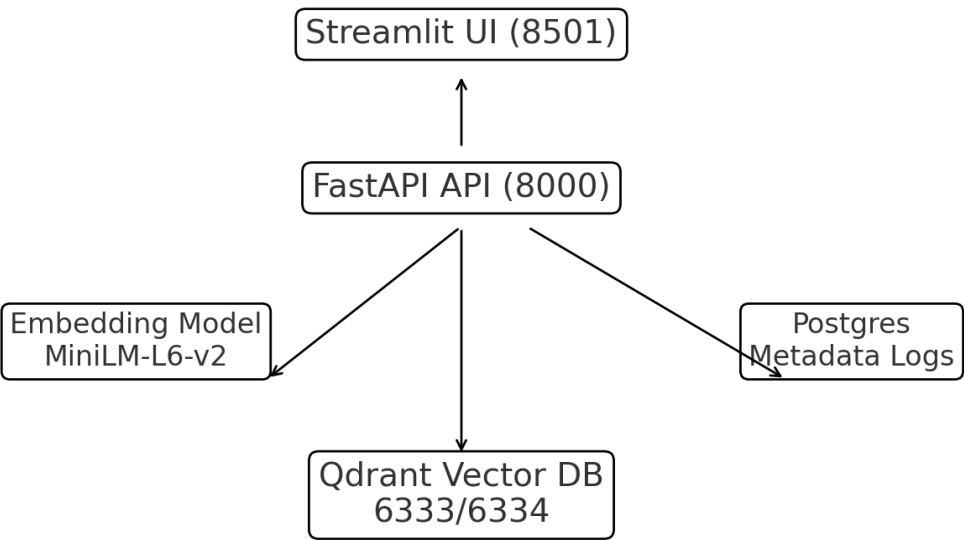


Enterprise RAG System - Documentation

System Architecture Diagram:



Full README Content:

```
# Enterprise RAG Application

This repository contains a modular, containerized Retrieval-Augmented Generation (RAG) system built
with FastAPI, Qdrant, Postgres, Streamlit, and Docker Compose.

---

## ■ System Architecture

### High-Level Architecture

![[Architecture Diagram]](architecture.png)

---
```

■ Project Structure

...

RAG-Application/

■ ■ ■ core/

■ ■ ■ storage/

■ ■ ■ retriever/

■ ■ ■ services/

■ ■ ■ utils/

■

■ ■ ■ services/

■ ■ ■ api/

■ ■ ■ ■ app/

■ ■ ■ ■ Dockerfile

■ ■ ■ ingest/

■ ■ ■ ■ ingest.py

■ ■ ■ ■ Dockerfile

■ ■ ■ ui/

■ ■ ■ app.py

■ ■ ■ Dockerfile

■

■ ■ ■ data/

■ ■ ■ docker-compose.yml

■ ■ ■ README.md

...

■ Components Overview

1. API Service (FastAPI)

Handles:

- Query processing
- Vector search calls to Qdrant
- Embedding generation
- Response construction

Accessible at:

...

<http://localhost:8000/docs>

...

2. Streamlit UI

A simple frontend to query the RAG engine.

Accessible at:

...

<http://localhost:8501>

...

3. Ingestion Service

Loads CSV/PDF/text files, embeds text chunks, and inserts into Qdrant + Postgres metadata tables.

Run manually (if needed):

...

`docker compose run --rm ingest`

...

4. Qdrant Vector DB

Used for similarity search.

Dashboard:

...

<http://localhost:6333/dashboard>

...

5. PostgreSQL

Stores metadata about documents.

■ Docker Compose Setup

Start Everything

...

`docker compose up -d`

...

Stop Everything

...

`docker compose down -v`

...

■ Ingest Data

Place files inside:

...

`data/`

...

Then run:

...

`docker compose run --rm ingest`

...

■ Test Query

Use UI:

...

`http://localhost:8501`

...

Or API:

...

`curl -X POST http://localhost:8000/ask -H "Content-Type: application/json" -d '{"query": "What is RAG?"}'`

...

■■ Diagrams

Application Overview

(See **architecture.png** included)

■ Credits

Designed and built with guidance from ChatGPT.